

Class of 2029

PPOM2- OMM Lab #31 Prep lecture

Balanced Ligamentous Tension (BLT)— Cervical, Thoracic, Lumbosacral, Fibula

August 10-14, 2026

To Shan Li, D.O.
Professor
tli05@nyit.edu

Session Objectives

At the conclusion of the session/course, the student should be able to:

1. Discuss the principles and goals of balanced ligamentous tension (BLT)
2. Differentiate between direct and indirect forces
3. Discuss some indications and contraindications for BLT
4. Discuss and correctly demonstrate BLT for the cervical spine, thoracic spine, and lumbosacral spine, and fibula (including interosseous membrane)
5. Correctly demonstrate the following: cervical, metatarsal, fibula head, and rib diagnosis; rib muscle energy technique (MET); cervical soft tissue technique and muscle energy technique; perform **soft tissue techniques** (ones that you were instructed to perform for any of the above regions)
6. Review Material from FOM Lab 6 and Lab 9

BLT: Principles & Basic Steps

- The basic principles behind BLT are “the minimization of peri-articular tissue load and the placement of the affected ligaments in a position of equal tension in all appropriate planes so that the body's inherent forces can resolve the somatic dysfunction.” The three basic steps for BLT are the following:
 - Disengage: Compress or decompress the tension of the tissues around a joint/segment to encourage *freeplay*
 - Exaggerate: Move in the direction of the documented dysfunctional pattern, which corresponds to the original position of injury, until a point of balance is determined.
 - Balance: Once the balance point has been achieved, maintain this position (point where tension in all directions is equal) until a release occurs.
- BLT can be performed indirectly or directly. In practice, it is performed indirectly most of the time.

Nicholas, Alexander S., Nicholas, Evan A., Atlas of Osteopathic Techniques, 4th Edition, Wolters Kluwer, 2023, Chapter 14: Balanced Ligamentous Tension and Ligamentous Articular Strain Techniques, see “General Information for All Dysfunctions” (**Treatment**) LINK:

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=252635966&bookid=3202#252635985>

About technique

- Forces
- Practice awareness, take the time
- Follow principles
 - BLT, though categorized as passive, can involve respiratory cooperation
 - As always with any OMT, reassess (with BLT, you are constantly assessing)

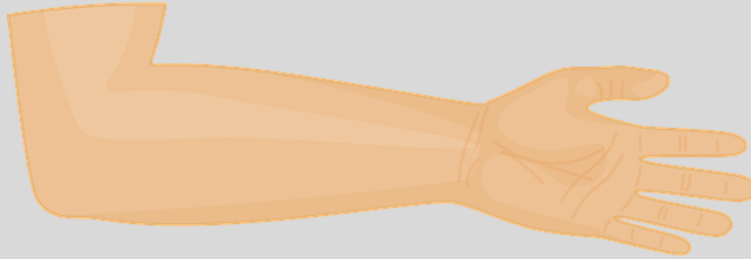
BLT Techniques taught in this lab:

- Cervical spine (C2-7)
- Thoracic spine (T1-12)
- Lumbosacral spine (L5-S1)
- Fibula (including interosseous membrane)

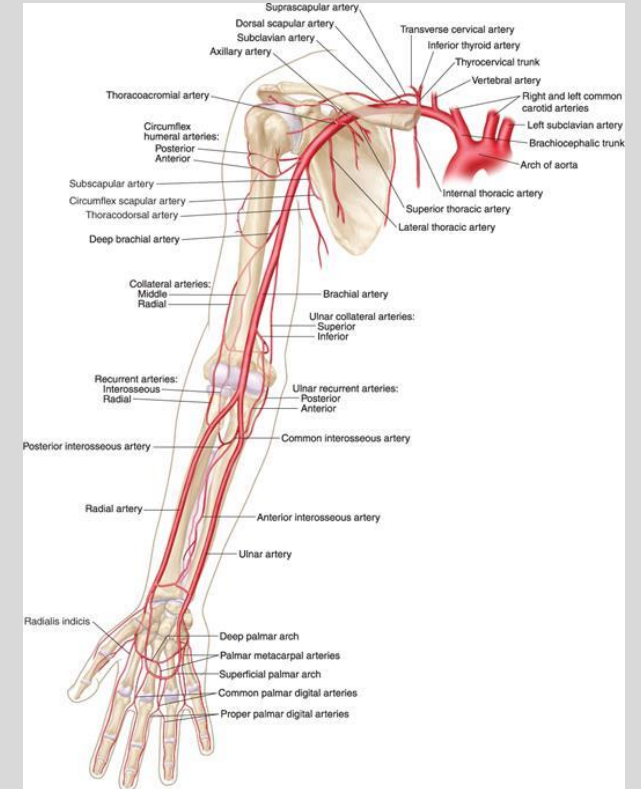
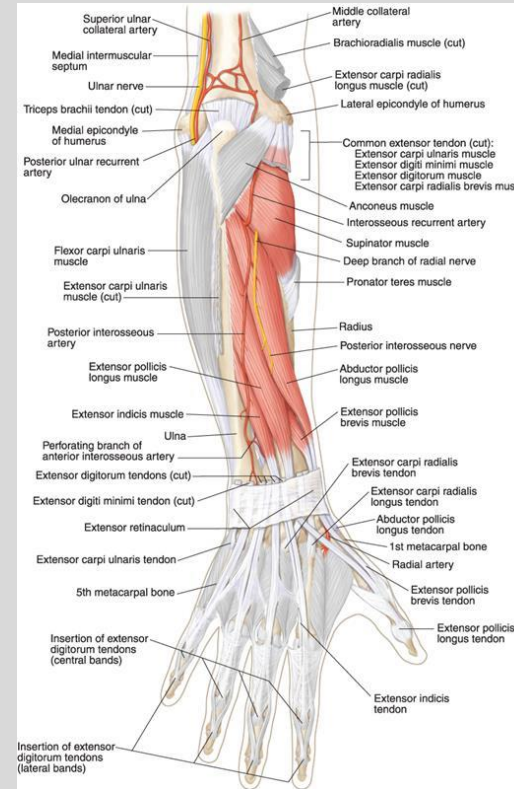


Palpatory Exercises

- Forearm



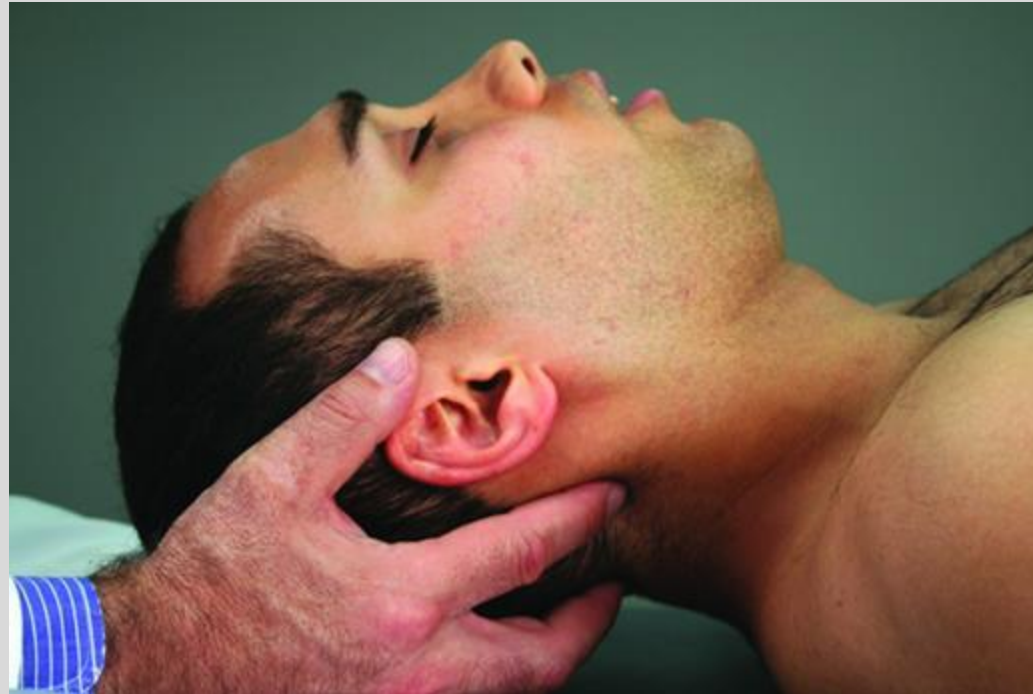
- Lumbosacral spine



Lippincott Atlas of Anatomy 2e., Chp. 2: The Upper Limb. Link:
<https://premiumbasicsscience-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=249368698&bookid=2991>

BLT of the Cervical Spine (C2-7)

Eg. C3 ERS right



Nicholas, Alexander S., Nicholas, Evan A., Atlas of Osteopathic Techniques, 4th Edition, Wolters Kluwer, 2023, Chapter 14: Balanced Ligamentous Tension and Ligamentous Articular Strain Techniques, see “General Information for All Dysfunctions” (**Cervical Region**); LINK (for picture): <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=252635966&bookid=3202#252636003>

BLT of the Thoracic Spine: T1-2 “Non-neutral” Type II Dysfunctions (eg. T1 FRS right)

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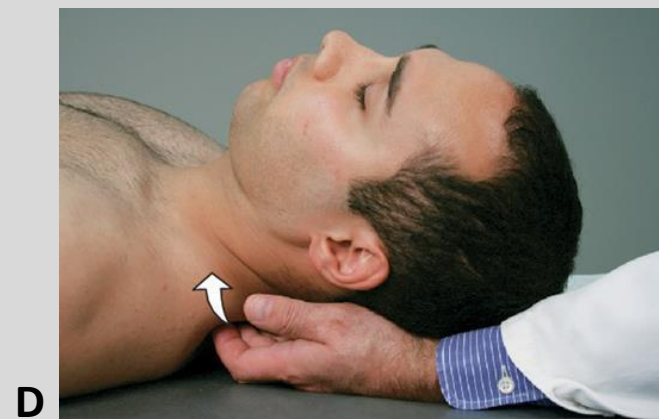
Contact T1&T2



Flexion-Extension of T2



SB & Rot T2



SB & Rot T1 (to fine tune)

Nicholas, Alexander S., Nicholas, Evan A., Atlas of Osteopathic Techniques, 4th Edition, Wolters Kluwer, 2023, Chapter 14: Balanced Ligamentous Tension and Ligamentous Articular Strain Techniques, see “General Information for All Dysfunctions” (**Thoracic Region**); LINK: <https://osteopathicmedicine-wwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=252635966&bookid=3202#252636003>

BLT of the Thoracic Spine: T4-12 “Neutral” Type I Dysfunctions (e.g. T8 NSIRr)



A Contact T8 left transverse process:
Note how fingers *cross* the midline



B Induce T8 left side bending by *pulling*
ribcage and the pt. *lowering* shoulder



C Lift finger on the T8 left transverse process
to induce right rotation; *flex* that same
finger to induce T8 left side bending

Nicholas, Alexander S., Nicholas, Evan A., Atlas of Osteopathic Techniques, 4th Edition, Wolters Kluwer, 2023, Chapter 14: Balanced Ligamentous Tension and Ligamentous Articular Strain Techniques, see “General Information for All Dysfunctions” (**Thoracic Region**); LINK: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=252635966&bookid=3202#252636003>

BLT of the Thoracic Spine: T4-12 “Neutral” Type I Dysfunctions (e.g. T6 NSrRI)



A Contact the T6 *left* transverse process with left thumb, and the T7 *right* transverse process with right thumb



B Instruct the patient to *flex* until T6 (upper of the two) starts to move, then *slightly extend* to find the *flexion/extension* balance point



C Instruct the patient to *tilt* right shoulder down (ie. lower) and *rotate* torso left to find balance point

Nicholas, Alexander S., Nicholas, Evan A., Atlas of Osteopathic Techniques, 4th Edition, Wolters Kluwer, 2023, Chapter 14: Balanced Ligamentous Tension and Ligamentous Articular Strain Techniques, see “General Information for All Dysfunctions” (**Thoracic Region**); LINK: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=252635966&bookid=3202#252636003>

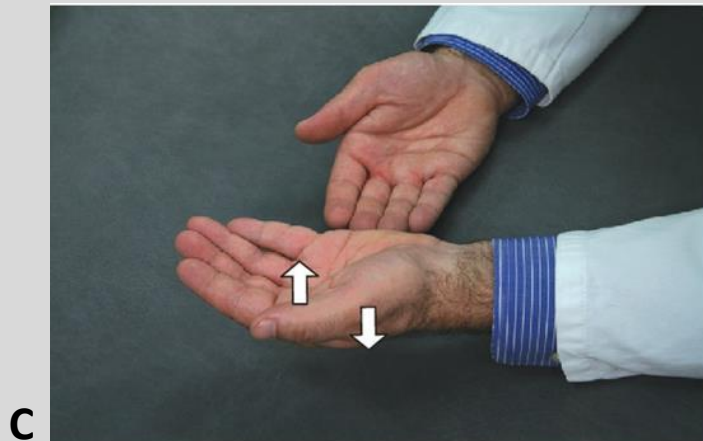
BLT of the Lumbosacral Spine: L5-S1 (lumbosacral) Dysfunction (e.g. L5 FRS right)



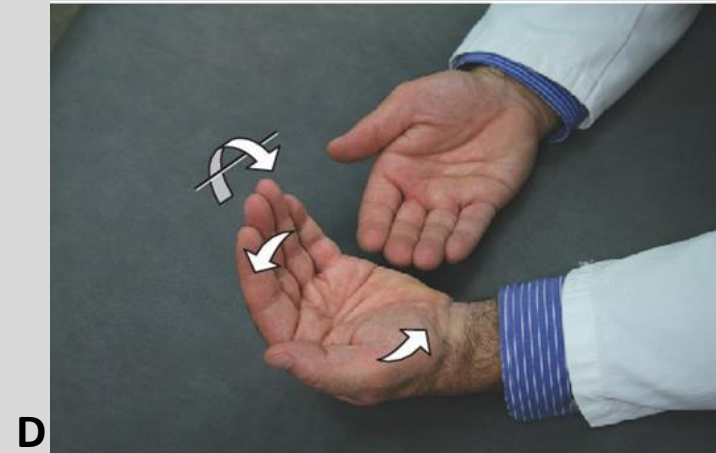
*Contact across
L5 and along
the sacrum*



Move each hand in
cranially-caudally
in direction of ease
to find balance
point between
lumbar and sacrum



*Lift or lower
hand to
balance
between
flexion-
extension*

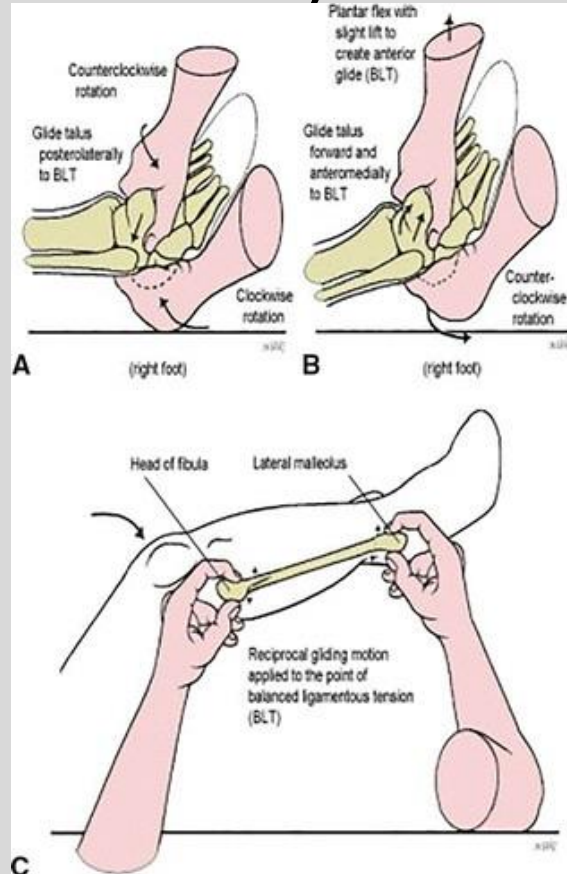


*Rotate and side bend
L5 to the right to
balance these
directions*

Nicholas, Alexander S., Nicholas, Evan A., Atlas of Osteopathic Techniques, 4th Edition, Wolters Kluwer, 2023, Chapter 14: Balanced Ligamentous Tension and Ligamentous Articular Strain Techniques, see "General Information for All Dysfunctions" (**Thoracic and Lumbar Regions**); LINK:

<https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=252635966&bookid=3202#252636003>

BLT of the Fibula (and interosseous membrane)



Seffinger, Michael A, Executive Editor, Foundations of Osteopathic Medicine, 4th Edition, Wolters Kluwer, 2018, Chapter 49C: "Osteopathic Approaches to Diagnosing and Treating Somatic Dysfunction in the Extremities": Lower Extremities" (OMT: BLT of Fibula and BMT of the Interosseous Membrane),
LINK: <https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=209559452&bookid=2582#209559670>

Thank you

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>